

Data processing

Notebook inputs
Raw EEG, event tables,
analysis-ready trial labels

Stimulus-locked preprocessing
band-pass 1–30 Hz
resample to 128 Hz
2 s epochs per trial

Epoch cache
trial x channel x time tensor
labels, groups, trial index

Feature engineering and model selection

Pre-modeling EEG inspection
participant class balance
waveforms, GFP, topographies

Pre-modeling figures
overview + MNE scalp maps

Feature engineering inside pipeline
CSP spatial filters
log-variance component features

Classifier
LDA with shrinkage

Grouped model selection
StratifiedGroupKFold
sweep CSP components
2, 4, 6, 8, 10, 12

Tuning outputs
best component count
grouped dummy vs model metrics

Modeling, evaluation, and reporting

Generalization stress test
Leave-one-subject-out
out-of-fold probabilities

Subject-level outputs
LOSO table
within-subject reference scores

Final notebook deliverables
summary figure + interpretation
cross-subject modellability

Prediction outputs
OOF prediction table
ROC, confusion, probability plots